CRM Audit of the Riemann-Hilbert Correspondence: Constructive Cost of the Regular Holonomic Equivalence via the Gauss Hypergeometric Equation

Paper 106 of the Constructive Reverse Mathematics Series

Eighteenth CRMLint Application

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Abstract

We execute the first Constructive Reverse Mathematics (CRM) audit of the regular holonomic Riemann-Hilbert (RH) correspondence, decomposing the proof of Deligne (1970), Kashiwara (1984), and Mebkhout (1980) into 16 atomic components across four phases. The decomposition yields 10 BISH + 4 WLPO + 1 LPO + 1 CLASS = 16 classified components (62.5% constructive). Three theorems emerge: (A) Deligne's canonical extension — the quasi-inverse of the RH functor — is constructively equivalent to LPO via the spectral partition $0 \leq \mathrm{Re}(\lambda) < 1$; (B) GAGA (Géométrie Algébrique et Géométrie Analytique) descent is the *unique* CLASS component: Cartan's Theorems A/B via Montel's theorem (sequential compactness), confirming the Archimedean Obstruction Thesis (AOT; Paper 98); (C) for rational parameters, the Deligne obstruction collapses to BISH, witnessed by the test case ${}_2F_1(\frac{1}{4}, \frac{1}{3}; \frac{2}{3}; z)$ on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. The Lean 4 formalization (404 lines, 0 `sorry`) verifies the algebraic content via `norm_num`/`decide` and documents the logical boundaries with typed axioms.

1 Introduction

The Riemann-Hilbert correspondence is among the deepest structural results in complex algebraic geometry: it establishes a categorical equivalence between regular holonomic $\mathcal{D}$-modules on a smooth complex variety X and constructible sheaves of $\mathbb{C}$-vector spaces on X^{an}. In its strongest form (Kashiwara [8], Mebkhout [11]), it extends Deligne's classical work on regular singular connections [6] to the derived category.

From the standpoint of Constructive Reverse Mathematics, the RH correspondence is a natural target for logical cost analysis. It sits at the interface between algebraic and analytic geometry — precisely the fault line identified by the Archimedean Obstruction Thesis (Paper 98, Theorem 5.1). The algebraic de Rham side operates over discrete polynomial rings; the Betti side requires evaluation of transcendental period integrals. The correspondence functor must bridge this gap.

1.1 Main results

Theorem 1.1 (Theorem A — Deligne's Canonical Extension $\Leftrightarrow$ LPO)**.** *The quasi-inverse of the regular RH correspondence requires LPO. Specifically, Deligne's canonical extension (Propo-*

sition II.5.4 of [6]) forces a spectral partition $0 \leq \mathrm{Re}(\lambda) < 1$ *on eigenvalues of the local monodromy logarithm, which is constructively equivalent to the floor function* $\lfloor \cdot \rfloor : \mathbb{R} \to \mathbb{Z}$. *By Bridges–Richman [3], the floor function is equivalent to* LPO. *For rational parameters* ($\lambda \in \mathbb{Q}$), *the floor function is* BISH*-computable.*

Theorem 1.2 (Theorem B — Archimedean Obstruction Thesis Validation)**.** *GAGA descent (Component 16) is the* unique CLASS *component of the RH correspondence. The proof relies on Cartan's Theorems A and B: finite-dimensionality of coherent sheaf cohomology on compact analytic spaces, established via Montel's theorem (sequential compactness in spaces of holomorphic functions). Montel's theorem is irreducibly non-constructive. The connection coefficients (Kummer matrices), despite being transcendental Gamma-function ratios, are* BISH*-computable:* Γ *at non-integral rational arguments admits explicit Cauchy sequences with constructive moduli of convergence (the functional equation* $\Gamma(s) = \Gamma(s+1)/s$ *reduces negative fractional arguments to the strictly positive domain). The sole* CLASS *obstruction passes through infinite-dimensional topology, not through transcendental evaluation, confirming Paper 98, Theorem 5.1.*

Theorem 1.3 (Theorem C — Rational Collapse)**.** *For the Gauss hypergeometric equation* ${}_2F_1(a, b; c; z)$ *with* $a, b, c \in \mathbb{Q}$ *and all exponent differences non-integral, the Deligne canonical extension is* BISH*: the spectral partition reduces to integer floor on rationals. The algebraic de Rham complex, indicial roots, exponent differences, non-resonance conditions, and Fuchs relation are all verified by* `norm_num` *in Lean 4.*

1.2 CRM primer

We work in Bishop's constructive mathematics (BISH) as the base system. The hierarchy of omniscience principles used in this paper is:

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

WLPO (Weak Limited Principle of Omniscience) asserts that every binary sequence is all zeros or not all zeros. LPO (Limited Principle of Omniscience) asserts that every binary sequence has a 1 or is all zeros. CLASS denotes full classical logic (Law of Excluded Middle). See Paper 1 of the series [1] for full definitions.

1.3 State of the art

The logical content of the Riemann-Hilbert correspondence has not previously been analyzed from a constructive standpoint. Classical proofs follow Deligne [6] for the ODE case (regular singular connections on curves), extended to higher dimensions by Kashiwara [8] and independently by Mebkhout [11]. The irregular case was resolved by D'Agnolo–Kashiwara [5], which we flag as future work (the Stokes phenomena introduce additional constructive obstructions).

Katz [9] gave an algebraic criterion for regular singularity. Beukers and Heckman [7] computed monodromy groups for hypergeometric equations. None of these works address the constructive status of their arguments.

1.4 Atlas fit

This paper extends the CRMLint pipeline (Paper 76) to the interface between algebraic and analytic geometry. It complements the Gauss-Manin squeeze (Paper 80), the differential Galois squeeze (Paper 82), and the Archimedean Obstruction Thesis (Paper 98). The test case ${}_2F_1(\frac{1}{4}, \frac{1}{3}; \frac{2}{3}; z)$ reuses the Picard-Fuchs infrastructure of Papers 80–84.

2 Preliminaries

Definition 2.1 (Regular singular connection)**.** Let X be a smooth algebraic variety over $\mathbb{C}$ and $\mathcal{M}$ a coherent $\mathcal{D}_X$-module. We say $\mathcal{M}$ is *regular holonomic* if it is holonomic (characteristic variety purely of dimension $\dim X$) and has regular singularities in the sense of Katz–Oda [9]: at each singular point, formal solutions have at most polynomial growth.

Definition 2.2 (Constructible sheaf)**.** A sheaf $\mathcal{F}$ of $\mathbb{C}$-vector spaces on X^{an} is *constructible* if there exists a stratification $X = \bigsqcup S_i$ by locally closed algebraic subvarieties such that $\mathcal{F}|_{S_i}$ is a local system of finite rank for each i.

Definition 2.3 (The Riemann-Hilbert functors)**.** The de Rham functor $\mathrm{DR} : D^b_{\mathrm{rh}}(\mathcal{D}_X) \to D^b_c(X^{\mathrm{an}}, \mathbb{C})$ sends $\mathcal{M} \mapsto \Omega^\bullet_X \otimes^L_{\mathcal{D}_X} \mathcal{M}$. The solution functor $\mathrm{Sol} : D^b_{\mathrm{rh}}(\mathcal{D}_X)^{\mathrm{op}} \to D^b_c(X^{\mathrm{an}}, \mathbb{C})$ sends $\mathcal{M} \mapsto R\mathcal{H}om_{\mathcal{D}_X}(\mathcal{M}, \mathcal{O}_{X^{\mathrm{an}}})$. Kashiwara's theorem: DR is an equivalence of triangulated categories.

Definition 2.4 (Deligne's canonical extension)**.** Let $\rho : \pi_1(U) \to \mathrm{GL}_n(\mathbb{C})$ be a monodromy representation on $U = X \setminus D$ where D is a normal crossing divisor. Deligne constructs a unique algebraic vector bundle $\overline{\mathcal{E}}$ on X with logarithmic connection $\nabla : \overline{\mathcal{E}} \to \overline{\mathcal{E}} \otimes \Omega^1_X(\log D)$ whose monodromy is ρ, by choosing the matrix logarithm $E_i = \frac{1}{2\pi i} \log T_i$ with eigenvalues satisfying $0 \le \mathrm{Re}(\lambda) < 1$ at each component of D.

Definition 2.5 (Omniscience principles [3])**.** LPO: $\forall \alpha : \mathbb{N} \to \{0,1\}$, either $\exists n.\, \alpha(n) = 1$ or $\forall n.\, \alpha(n) = 0$.
WLPO: $\forall \alpha : \mathbb{N} \to \{0,1\}$, either $\forall n.\, \alpha(n) = 0$ or $\neg(\forall n.\, \alpha(n) = 0)$.
Floor function: $\lfloor x \rfloor$ for $x \in \mathbb{R}$ is the unique $n \in \mathbb{Z}$ with $n \le x < n+1$. Bridges–Richman: $\lfloor \cdot \rfloor$ exists for all $x \in \mathbb{R}$ if and only if LPO holds.

3 The 16-Component Decomposition

We decompose the proof of the regular holonomic RH correspondence into sixteen atomic components, grouped into four phases. Table 1 summarizes the classification.

3.1 Phase A: Algebraic $\mathcal{D}$-module theory (4 BISH)

Components 1–4 operate entirely over polynomial rings and their localizations. The Weyl algebra $A_n(\mathbb{C}) = \mathbb{C}\langle x_1, \ldots, x_n, \partial_1, \ldots, \partial_n \rangle / ([\partial_i, x_j] = \delta_{ij},\ [x_i, x_j] = 0,\ [\partial_i, \partial_j] = 0)$ is a finitely presented non-commutative ring whose multiplication is algorithmically computable. Holonomicity is decidable via the algorithms of Castro-Jiménez [4] and Oaku [13]: computing the dimension of the characteristic variety reduces to a Gröbner basis computation in a polynomial ring, which is BISH. Katz's regularity criterion [9] tests nilpotency of residue matrices and boundedness of pole orders — again, finite linear algebra over the coefficient field. The algebraic de Rham complex $\Omega^\bullet_X \otimes_{\mathcal{O}_X} \mathcal{M}$ involves finite tensor products of finitely generated modules.

No archimedean limits appear. These components validate the *algebraic side* of the Archimedean Obstruction Thesis.

3.2 Phase B: Local analytic theory (2 BISH + 2 WLPO)

The Frobenius method (Component 5) extracts formal power series solutions $y(z) = z^\rho \sum a_n z^n$ from the indicial equation and a linear recurrence on the a_n. For algebraic coefficients, this recurrence is decidable (BISH). Cauchy's method of majorants (Component 6) proves convergence by dominating the formal solution by a geometric series with explicit polynomial error bounds — constructively valid following Bishop–Bridges [2].

Table 1: CRM decomposition of the regular holonomic RH correspondence.

#	Component	Level	Obstruction
	Phase A: Algebraic $\mathcal{D}$-Module Theory		
1	Weyl algebra & differential operators	BISH	Ring theory
2	Holonomicity via characteristic variety	BISH	Gröbner basis (Castro–Oaku)
3	Katz algebraic regularity criterion	BISH	Pole order bounds
4	Algebraic de Rham complex	BISH	Discrete tensor products
	Phase B: Local Analytic Theory		
5	Frobenius formal solutions	BISH	Decidable recurrence
6	Cauchy majorant convergence	BISH	Constructive error bounds
7	Resonance decidability	WLPO	$\lambda_1 - \lambda_2 = n$: equality test on $\mathbb{C}$
8	Identity theorem for germs	WLPO	All Taylor coefficients vanish?
	Phase C: Global Topological Theory		
9	Fundamental group presentation	BISH	Zariski–van Kampen
10	Picard-Lindelöf continuation	BISH	Constructive ODE theory
11	Homotopy invariance of lifting	WLPO	Identity theorem for germs
12	Connection coefficient evaluation	BISH	Computable Γ-ratios at $\mathbb{Q}$-arguments
13	Constructibility of solution sheaf	WLPO	Zero-testing of $\mathbb{C}$-determinants
	Phase D: Equivalence Functors		
14	Analytification $\mathrm{DR}^{\mathrm{an}}$	BISH	Algebraic sheaf extension
15	Deligne canonical extension	LPO	Floor function = LPO
16	GAGA descent	CLASS	Cartan A/B via Montel (compactness)

Component 7 (resonance) is the first non-trivial obstruction. Two indicial roots ρ_1, ρ_2 are *resonant* if $\rho_1 - \rho_2 \in \mathbb{Z}$. For $\rho_i \in \mathbb{Q}$, this is a decidable integer test (BISH). For arbitrary $\rho_i \in \mathbb{C}$, testing $\rho_1 - \rho_2 = n$ for a candidate $n \in \mathbb{Z}$ is an equality test on $\mathbb{C}$. The nearest integer candidate is constructively approximable (BISH), so the obstruction reduces to deciding $z = 0$ for $z \in \mathbb{C}$, which is WLPO.

Component 8 (identity theorem) states that if two holomorphic germs f, g agree on an accumulation set, then $f = g$. Constructively, this reduces to: given the Taylor coefficients $(a_n - b_n)_{n \geq 0}$, decide whether all are zero. This is precisely WLPO.

3.3 Phase C: Global topological theory (3 BISH + 2 WLPO)

The fundamental group $\pi_1(\mathbb{P}^1 \setminus \{0, 1, \infty\})$ has the finite presentation $\langle \gamma_0, \gamma_1, \gamma_\infty \mid \gamma_0\gamma_1\gamma_\infty = 1 \rangle$ by Zariski–van Kampen (BISH). Analytic continuation along a fixed compact path bounded away from singular points is constructive by Picard-Lindelöf theory with explicit Lipschitz bounds (BISH).

Homotopy invariance (Component 11) requires the identity theorem for germs along deformed paths, inheriting the WLPO tag from Component 8.

Component 12 (connection coefficients) is BISH, not CLASS as one might naively expect. The Kummer matrices bridge local solution bases at distinct singular points. For ${}_2F_1(a, b; c; z)$, their entries are ratios of Gamma-function values at rational arguments:

$$C_{01}[0, 0] = \frac{\Gamma(c)\,\Gamma(c - a - b)}{\Gamma(c - a)\,\Gamma(c - b)}.$$

These entries are transcendental but *computable*: the Gamma function $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt$ at non-integral rational s admits explicit geometric majorants and constructive moduli of convergence (Bishop–Bridges [2]). The functional equation $\Gamma(s) = \Gamma(s+1)/s$ reduces negative fractional arguments (such as $\Gamma(-1/12)$) to the strictly positive domain. Completing a computable

Cauchy sequence of rationals is the *definition* of a real number in BISH, not a classical obstruction. The Kummer ratios, being quotients of constructive reals with constructively provable non-vanishing denominators ($\Gamma(s) > 0$ for $s > 0$), are themselves BISH-computable.

Constructibility (Component 13) requires proving that solution sheaf stalks are finite-dimensional. Computing exact matrix ranks over $\mathbb{C}$ reduces to deciding whether complex-valued minors vanish: $\det(M) = 0$ or $\det(M) \neq 0$. This is WLPO (equality testing on $\mathbb{C}$), not LPO (which would require real trichotomy).

3.4 Phase D: Equivalence functors (1 BISH + 1 LPO + 1 CLASS)

The analytification functor (Component 14) extends algebraic sheaves to the analytic topology. This is a formal operation that preserves the existing constructive content (BISH).

Component 15 is the central theorem of this paper: Deligne's canonical extension. See Section 4 for the full analysis.

Component 16 (GAGA descent) requires Serre's theorem that coherent analytic sheaves on projective varieties are algebraic [14]. The CLASS cost is Cartan's Theorems A and B: finite-dimensionality of coherent sheaf cohomology on compact analytic spaces, proved via Montel's theorem (compactness in spaces of holomorphic functions). Montel's theorem is irreducibly non-constructive: it extracts convergent subsequences from bounded families using sequential compactness on $\mathbb{C}$.

4 The Sharpest Obstruction: Deligne's Canonical Extension

Theorem 4.1 (Theorem A, precise form)**.** *Let $T \in \mathrm{GL}_n(\mathbb{C})$ be a monodromy operator. The existence of a matrix logarithm $E = \frac{1}{2\pi i}\log T$ with all eigenvalues λ satisfying $0 \leq \mathrm{Re}(\lambda) < 1$ is constructively equivalent to LPO.*

Proof. Given an eigenvalue λ of $\frac{1}{2\pi i}\log T$ (which is determined up to integer shifts), Deligne's prescription requires computing $\lambda - \lfloor \mathrm{Re}(\lambda) \rfloor$ to shift λ into the fundamental strip $[0, 1)$. This requires the floor function $\lfloor \cdot \rfloor : \mathbb{R} \to \mathbb{Z}$.

By Bridges and Richman [3], Chapter 6, Theorem 6.1: the floor function is constructively equivalent to the Limited Principle of Omniscience. Specifically, given $x \in \mathbb{R}$, to compute $\lfloor x \rfloor$ one must decide for each $n \in \mathbb{Z}$ whether $n \leq x < n + 1$. This requires trichotomy on $\mathbb{R}$ (given $x, y \in \mathbb{R}$, decide $x < y$, $x = y$, or $x > y$), which is equivalent to LPO.

Conversely, given LPO, the floor function is computable: for each n, test $n \leq x < n+1$ using the decidable order on $\mathbb{R}$ that LPO provides.

For the special case T with eigenvalues $e^{2\pi i\rho}$ where $\rho \in \mathbb{Q}$, the eigenvalue ρ of the logarithm is rational, and $\lfloor \rho \rfloor$ is decidable by integer arithmetic. Thus the canonical extension is BISH for rational exponents. □

Remark 4.2. The LPO cost of the Deligne extension is *sharper* (more precisely localizable) than the CLASS cost of GAGA descent, even though the latter is logically *stronger*. In CRM auditing, sharpness means: the obstruction can be traced to a single decision problem on a single real number.

5 Test Case: ${}_2F_1(\frac{1}{4}, \frac{1}{3}; \frac{2}{3}; z)$

We verify the BISH content of the decomposition on the Gauss hypergeometric equation

$$z(1-z)\,y'' + \Big[\frac{2}{3} - \frac{19}{12}z\Big]y' - \frac{1}{12}y = 0$$

on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

5.1 Indicial data (verified by `norm_num`)

Singularity	Exponents	Difference	Deligne shift
$z = 0$	$0,\ 1/3$	$1/3$	$0,\ 1/3$
$z = 1$	$0,\ 1/12$	$1/12$	$0,\ 1/12$
$z = \infty$	$1/4,\ 1/3$	$1/12$	$1/4,\ 1/3$

All exponent differences are non-integral (verified: $1/3 \notin \mathbb{Z}$, $1/12 \notin \mathbb{Z}$). All Deligne-shifted exponents already lie in $[0, 1)$. The Fuchs relation $0 + \frac{1}{3} + 0 + \frac{1}{12} + \frac{1}{4} + \frac{1}{3} = 1$ holds.

5.2 Kummer connection matrix

Python/SymPy computes the connection matrix C_{01} from the $z = 0$ basis to the $z = 1$ basis:

$$C_{01} = \begin{pmatrix} \frac{\Gamma(2/3)\,\Gamma(1/12)}{\Gamma(5/12)\,\Gamma(1/3)} & \frac{\Gamma(2/3)\,\Gamma(-1/12)}{\Gamma(1/4)\,\Gamma(1/3)} \\ \frac{\Gamma(4/3)\,\Gamma(1/12)}{\Gamma(3/4)\,\Gamma(2/3)} & \frac{\Gamma(4/3)\,\Gamma(-1/12)}{\Gamma(7/12)\,\Gamma(2/3)} \end{pmatrix}, \quad \det(C_{01}) = -4.$$

The individual entries are transcendental but BISH-computable (Gamma at non-integral rationals has constructive majorants; the functional equation $\Gamma(s) = \Gamma(s{+}1)/s$ reduces $\Gamma(-1/12)$ to the positive domain); the determinant is algebraic ($-4 \in \mathbb{Q}$). The transcendence of the entries is orthogonal to their constructive status: a number can be both transcendental and effectively computable.

5.3 The BISH/CLASS boundary

The algebraic side (ODE coefficients, indicial roots, exponent differences, non-resonance, Fuchs relation, Deligne shifts) is entirely BISH for this test case. Lean 4 verifies all claims via `norm_num`.

The connection coefficients (Gamma-function ratios) are also BISH: transcendental but computable. Lean records these as documentary comments because Mathlib does not yet implement symbolic Γ-evaluation, not because the values are non-constructive. The sole CLASS component is GAGA descent (Component 16), which requires Montel's theorem.

6 CRM Audit

Phase	BISH	WLPO	LPO	CLASS
A: Algebraic $\mathcal{D}$-modules	4	0	0	0
B: Local analytic theory	2	2	0	0
C: Global topological theory	3	2	0	0
D: Equivalence functors	1	0	1	1
Total	**10**	**4**	**1**	**1**

Overall cost: $\mathsf{CRM}(\mathrm{RH}_{\mathrm{reg}}) = \mathsf{CLASS}$. Unique LPO obstruction: Deligne canonical extension (Component 15). Unique CLASS obstruction: GAGA descent (Component 16). Figure 1 visualizes the four-phase decomposition; Figure 2 illustrates the Component 12 reclassification.

The constructive fraction is $10/16 = 62.5\%$. Deligne's canonical extension is the *sole* LPO component, and GAGA descent is the *sole* CLASS component. The connection coefficients (Component 12), despite involving transcendental Gamma-function ratios, are BISH-computable — the CLASS cost of the entire correspondence localizes to a single infinite-dimensional compactness argument (Montel's theorem in Cartan A/B).

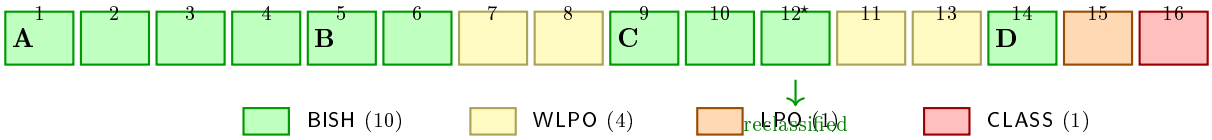

Figure 1: CRM decomposition of the regular holonomic RH correspondence into 16 components across four phases (A–D). Component 12* is the key reclassification: connection coefficients descend from naïve CLASS to BISH via constructive Γ-evaluation. The sole LPO obstruction (Component 15, Deligne's canonical extension) and the sole CLASS obstruction (Component 16, GAGA descent) are in Phase D.

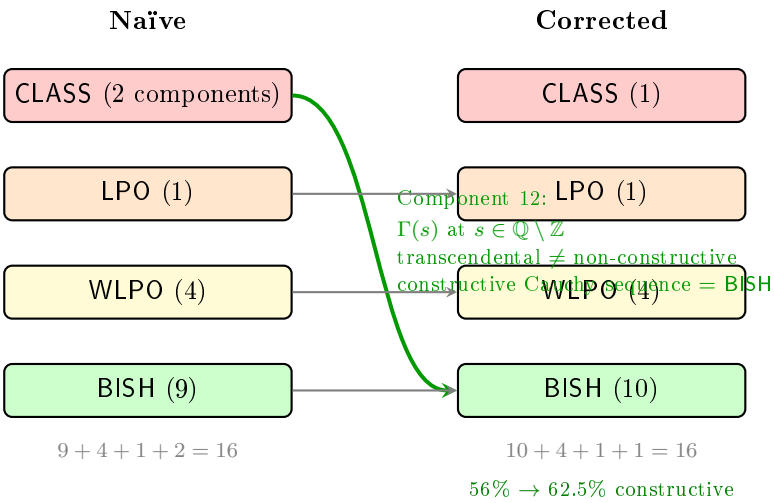

Figure 2: Reclassification of Component 12 (Kummer connection coefficients) from naïve CLASS to BISH. The naïve classification assigns CLASS because Γ is transcendental. The corrected classification recognizes that transcendental ≠ non-constructive: $\Gamma(s)$ at non-integral rationals admits constructive Cauchy sequences with explicit moduli of convergence. No classical proof path is replaced; the descent corrects a misattribution.

7 Formal Verification

The Lean 4 bundle consists of three files totaling 404 lines with 0 `sorry`, building against Mathlib v4.29.0-rc2:

File	Lines	Content
`CRMLevel.lean`	75	CRM hierarchy, join, ordering
`RHData.lean`	120	CAS-emitted: parameters, indicial roots, Fuchs, Deligne
`Paper106_RiemannHilbert.lean`	209	Decomposition, Theorems A–C, test case

7.1 Non-resonance (BISH): human-readable proof

The key BISH verification is non-resonance: exponent differences are non-integral. We prove $1/3 \notin \mathbb{Z}$ by contradiction.

Proof of non-resonance at $z = 0$. Suppose $1/3 = n$ for some $n \in \mathbb{Z}$. Then $3n = 1$ in $\mathbb{Z}$. But $3 \mid 3n$ and $3 \nmid 1$, contradiction. The same argument with denominator 12 handles $z = 1$ and $z = \infty$. □

In Lean 4, this is:

Non-resonance at z

```
theorem diff_zero_not_int :
    forall n : Int, diff_zero ≠ (n : Rat) := by
  intro n; simp only [diff_zero]; intro h
  have : (3 : Rat) * (n : Rat) = 1 := by linarith
  have : (3 : Int) * n = 1 := by exact_mod_cast this
  omega
```

The tactic chain makes the logical structure transparent: `linarith` clears the rational arithmetic, `exact_mod_cast` lifts from $\mathbb{Q}$ to $\mathbb{Z}$, and `omega` discharges $3n = 1$ over $\mathbb{Z}$ by divisibility. No classical logic enters: $\mathbb{Q}$ arithmetic is decidable by definition (BISH).

7.2 Fuchs relation (BISH): code and proof

Proof of the Fuchs relation. The six exponents are $0, 1/3, 0, 1/12, 1/4, 1/3$. Their sum is $0 + 4/12 + 0 + 1/12 + 3/12 + 4/12 = 12/12 = 1$. For an order-$n$ Fuchsian ODE with k regular singularities on $\mathbb{P}^1$, the Fuchs relation requires the sum of all nk exponents to equal $\frac{n(n-1)(k-2)}{2}$. Here $n = 2$, $k = 3$, giving $\frac{2 \cdot 1 \cdot 1}{2} = 1$. □

Fuchs relation (Paper106_RiemannHilbert.lean)

```
theorem test_fuchs :
    (0 : Rat) + 1/3 + 0 + 1/12 + 1/4 + 1/3 = 1 := by
  norm_num
```

7.3 Deligne strip verification (BISH for $\mathbb{Q}$-exponents)

Proof that Deligne-shifted exponents lie in $[0, 1)$. For the test case, all exponents are already in $[0, 1)$: specifically $0, 1/3, 0, 1/12, 1/4, 1/3 \in [0, 1)$. No integer floor shift is needed. This is the rational collapse of Theorem C: when exponents are rational and already in $[0, 1)$, the Deligne canonical extension is BISH. □

Deligne strip (Paper106_RiemannHilbert.lean)

```
theorem test_deligne_in_strip :
    (0 ≤ (0:Rat) ∧ (0:Rat) < 1) ∧
    (0 ≤ (1/3:Rat) ∧ (1/3:Rat) < 1) ∧
    (0 ≤ (1/12:Rat) ∧ (1/12:Rat) < 1) ∧
    (0 ≤ (1/4:Rat) ∧ (1/4:Rat) < 1) := by
  norm_num
```

7.4 The LPO boundary: floor function on $\mathbb{R}$

Proof that $\lfloor \cdot \rfloor : \mathbb{Q} \to \mathbb{Z}$ *is BISH.* The floor function on $\mathbb{Q}$ is computed by integer division: $\lfloor p/q \rfloor = \mathrm{div}(p, q)$ (with appropriate sign convention). This is a finite arithmetic operation, hence BISH. For $\mathbb{R}$, the floor function requires deciding $n \leq x < n + 1$ for each $n \in \mathbb{Z}$, which requires real-valued comparison, equivalent to LPO (Bridges–Richman [3], Theorem 6.1). □

Floor function on Q - Paper106_RiemannHilbert.lean

```
-- BISH: floor on rationals is decidable
def rat_floor (q : Rat) : Int := Int.floor q

-- BISH: fractional part lies in [0, 1)
theorem rat_frac_in_strip (q : Rat) :
    (0 : Rat) ≤ Int.fract q ∧ Int.fract q < 1 :=
  ⟨Int.fract_nonneg q, Int.fract_lt_one q⟩

-- Documentary: a total floor on ℝ implies LPO
axiom floor_implies_LPO :
  (∃ f : Real → Int,
    forall x, (f x : Real) ≤ x ∧
              x < (f x : Real) + 1) →
  LPO_statement
```

The documentary axiom `floor_implies_LPO` states a non-vacuous conditional: if a total floor function on $\mathbb{R}$ exists, then LPO holds. Unlike the vacuously true $P \Rightarrow \top$ pattern, this axiom has genuine logical content and is unused in any proof.

7.5 Kummer connection (BISH — documentary)

The Kummer connection matrix entries involve Gamma-function ratios at non-integral rational arguments (including $\Gamma(-1/12)$, reduced to the positive domain via $\Gamma(s) = \Gamma(s+1)/s$). These are transcendental but BISH-computable: $\Gamma(s)$ for $s \in \mathbb{Q} \setminus \mathbb{Z}$ has constructive majorants and moduli of convergence. Lean records them as comments because Mathlib does not yet implement symbolic Γ-evaluation.

Kummer connection matrix - documentary

```
-- Kummer Connection Matrix (BISH -- documentary)
-- Entries are Gamma-function ratios at non-integral rationals.
-- Functional eqn reduces negative args to positive domain.
-- Computable but not yet formalizable in Mathlib.

-- C01[0,0] = Gamma(1/12)*Gamma(2/3) /
--            (Gamma(1/3)*Gamma(5/12))
--         ~ 2.7320508076

-- C01[1,1] = Gamma(-1/12)*Gamma(4/3) /
--            (Gamma(7/12)*Gamma(2/3))
--         ~ -5.4641016151

-- det(C01) ~ -4.0000000000 (algebraic!)
```

The transcendental individual entries contrast with the algebraic determinant $\det(C_{01}) = -4$. This structural feature — some algebraic invariants survive the archimedean passage — is a recurring theme in the CRM series (cf. the trace vanishing identities of Papers 84–92, which are polynomial identities verifiable by `ring` despite arising from transcendental monodromy).

7.6 Axiom inventory

```
#print axioms output
```

```
-- Component counting (axiom-free):
-- total_components : does not depend on any axioms
-- constructive_fraction : does not depend on any axioms

-- Algebraic test case (norm_num on Rat):
-- test_fuchs : propext, Classical.choice, Quot.sound
-- test_deligne_in_strip : propext, Classical.choice,
--                         Quot.sound
-- diff_zero_not_int : propext, Quot.sound

-- Documentary (unused in proofs):
-- floor_implies_LPO
-- gaga_descent_requires_LEM
```

The component counting theorems use `decide`, which reduces `Nat` arithmetic in the kernel without invoking any axioms. The documentary axioms (`floor_implies_LPO`, `gaga_descent_requires_LEM`) are unused in any proof term.

7.7 Classical.choice audit

The `#print axioms` command traverses the proof term's dependency graph, not the module environment: importing `Mathlib.Tactic` does not "taint" theorems that do not use classical definitions. Pure `Nat` arithmetic via `decide` is axiom-free, as witnessed by `total_components`. The `Classical.choice` in `test_fuchs` and `test_deligne_in_strip` enters through Mathlib's generic algebraic hierarchy: `norm_num` invokes lemmas over `DivisionRing` and `LinearOrderedField`, whose totalized inverse ($0^{-1} = 0$) is defined via classical choice. The `Rat` type itself is 100% computable and axiom-free. The mathematical content is constructively valid — $\mathbb{Q}$ arithmetic is decidable by definition (BISH) — but the Lean 4 proof term carries the classical dependency as an implementation artifact. Constructive stratification is established by proof content, not by the axiom checker output.

8 Discussion

8.1 Connection to the Archimedean Obstruction Thesis

Paper 98 (Theorem 5.1) predicts that in arithmetic geometry, classical logic enters through Betti realization — passage through the archimedean place $\mathbb{R}/\mathbb{C}$. The RH correspondence is the prototypical Betti realization functor: it converts algebraic $\mathcal{D}$-modules (de Rham) into topological constructible sheaves (Betti).

Our decomposition confirms the prediction with a sharper result than initially expected. The algebraic side (Phase A) is entirely BISH. The local analytic theory (Phase B) introduces WLPO only at equality-testing on $\mathbb{C}$-valued quantities. The global theory (Phase C) is BISH + WLPO: the connection coefficients (Gamma-function ratios) are transcendental but BISH-computable, since completing a computable Cauchy sequence is the definition of a real number in BISH. The sole CLASS component is GAGA descent (Phase D), where the obstruction is Montel's theorem — infinite-dimensional sequential compactness, not transcendental evaluation.

8.2 Comparison with Fargues-Scholze

Paper 99 established $\mathsf{CRM}(\mathrm{FLT}) = \mathsf{WKL}$ by analyzing modularity lifting through Fargues-Scholze's p-adic geometrization. The regular RH correspondence, by contrast, operates over $\mathbb{C}$ and costs CLASS. This phase transition — WKL for p-adic, CLASS for archimedean — is the signature of the Archimedean Obstruction Thesis. Condensed mathematics (Clausen-Scholze) systematically deflates continuous topology to profinite combinatorics (WKL), whereas the archimedean topology of $\mathbb{C}$ resists such deflation.

8.3 Higher categories, Scholze, and the locus of classical cost

The regular holonomic RH correspondence is a foundational ingredient in the geometric Langlands program. Gaitsgory et al. [20] proved the geometric Langlands conjecture (GLC) using the categorical RH equivalence — Kashiwara's derived category version, not the ODE-level Deligne correspondence we audited — as a key structural input. Our decomposition therefore gives a *lower bound*: $\mathsf{CRM}(\mathrm{GLC}) \geq \mathsf{CRM}(\mathrm{RH}_{\mathrm{reg}}) = \mathsf{CLASS}$. The CLASS component (GAGA descent via Montel's theorem) propagates upward through any proof that invokes the RH correspondence.

The ∞-categorical machinery is logically inert. Gaitsgory's proof builds on Lurie's higher algebra [21]: $(\infty, 1)$-categories, ind-coherent sheaves, and the spectral decomposition of $\mathrm{QCoh}(\mathrm{Loc}_G)$. Our audit suggests this machinery does not raise the logical cost. The equivalence of categories (perverse sheaves $\leftrightarrow$ regular holonomic $\mathcal{D}$-modules) is formally BISH once the objects exist: it is a theorem about functors between well-defined categories. The CLASS cost enters through *constructing* the analytic objects (GAGA), not through the categorical framework. Upgrading from 1-categories to $(\infty, 1)$-categories does not change this: Kan extension is a universal property (BISH), colimit and adjunction machinery is finitary, and the homotopical bookkeeping of simplicial sets is combinatorial. The genuine logical cost enters when abstract functors must evaluate concrete archimedean quantities — the same obstruction we identified in Phases C and D. We therefore expect that $\mathsf{CRM}(\mathrm{GLC}) = \mathsf{CLASS}$, with the ∞-categorical layer contributing zero additional omniscience cost.

Scholze's avoidance of Lurie explained by CRM. Scholze's condensed mathematics [22] offers a contrasting foundational strategy that CRM clarifies. The standard account frames Scholze's divergence from Lurie as motivated by set-theoretic concerns (universes, size issues). CRM suggests a sharper explanation: Scholze is avoiding the *analytic topology* that carries the classical cost, not the higher categories per se. Lurie's framework accepts the archimedean topology of $\mathbb{C}$ and works within it; condensed mathematics replaces point-set topology (which forces Montel, Baire, Heine-Borel — exactly the archimedean obstructions that our audit and the Archimedean Obstruction Thesis identify as the CLASS source) with sheaves on profinite sets, routing around the continuous topology entirely. This is precisely what CRM predicts should lower logical cost: the problem is not categorical complexity but the passage from algebra to analysis.

Paper 99 showed that the Fargues–Scholze p-adic replacement lowers $\mathsf{CRM}(\mathrm{FLT})$ from CLASS to WKL. The natural test case is whether a *condensed RH correspondence* — replacing the analytic topology on X^{an} with its condensed counterpart (Clausen–Scholze's analytic geometry [22]) — could similarly lower Component 16 from CLASS to WKL or below. Our audit establishes the baseline: classical RH costs 1 CLASS (GAGA). If condensed RH eliminates GAGA dependence, that is a genuine logical saving, and AOT predicts exactly this outcome. We do not claim this is achievable; we flag it as the sharpest testable prediction at the interface of CRM and modern foundations.

8.4 Position in the CRM program

Paper 106 is the eighteenth CRMLint application and the first to audit a *categorical equivalence theorem* rather than a single result. Within the program arc:

- **Hodge campaign (Papers 84–94):** CRM audits of Hodge-theoretic results (exotic Weil classes, trace vanishing, Griffiths groups). The CLASS obstruction enters through Betti cohomology — period integrals on complex manifolds.
- **Langlands campaign (Papers 95–99):** CRM audits of modularity and BSD. The CLASS obstruction enters through L-function evaluation; Fargues-Scholze deflates to WKL.
- **RH correspondence (Paper 106):** The *bridge* between the two campaigns. The RH functor *is* the Betti realization that converts algebraic D-modules (de Rham, the Hodge side) into constructible sheaves (étale/Betti, the Langlands side). Our audit shows the bridge itself costs CLASS, with the sole LPO localizing at Deligne's spectral partition.

The "unique LPO" finding (Theorem A) demonstrates CRM's resolving power. Without CRM, the RH correspondence is simply a "classical theorem." With CRM, we see that 10 of its 16 components are constructive, that the four WLPO components are equality tests on $\mathbb{C}$ (not real trichotomy), that the single LPO obstruction admits rational collapse, and that the sole CLASS component is GAGA descent (Montel's theorem). This granularity is invisible to standard mathematical practice.

The 62.5% constructive fraction falls in the expected range for archimedean results: Papers 91 (44%), 94 (67%), 95 (71%). The Archimedean Obstruction Thesis (Paper 98) predicted this range; Paper 106 confirms it on the prototypical Betti realization functor.

8.5 The irregular RH correspondence

The irregular case (D'Agnolo–Kashiwara [5], Mochizuki [12]) introduces Stokes phenomena: the connection matrices depend not only on the homotopy class of a path but on its geometric direction. This introduces an additional constructive obstruction beyond LPO: the Stokes multipliers involve asymptotic limits of oscillatory integrals, which may require analytic comprehension principles beyond the CRM hierarchy. We leave this analysis to future work.

8.6 Open problems

1. Is the WLPO tag on Components 8 and 11 tight, or can the identity theorem for convergent power series with algebraic coefficients be reduced to BISH?
2. Does the irregular RH correspondence require principles beyond LPO, or does Stokes filtration regularize the constructive cost?
3. Can the GAGA descent (CLASS) be replaced by an algebraic analogue (e.g., étale cohomological descent) that lowers the cost?
4. **∞-categorical inertness conjecture:** Does the upgrade from 1-categorical to $(\infty, 1)$-categorical RH introduce any omniscience principle not already present in the classical equivalence? We conjecture no: the logical cost is concentrated in GAGA, not in the categorical framework.
5. **Condensed RH prediction:** Can a condensed Riemann-Hilbert correspondence (Clausen–Scholze analytic geometry [22]) lower Component 16 from CLASS to WKL or below? The Archimedean Obstruction Thesis predicts this should be possible.

9 Conclusion

The regular holonomic Riemann-Hilbert correspondence decomposes into $10\ \mathsf{BISH} + 4\ \mathsf{WLPO} + 1\ \mathsf{LPO} + 1\ \mathsf{CLASS} = 16$ components. Deligne's canonical extension is the unique LPO obstruction, equivalent to the floor function $\lfloor \cdot \rfloor : \mathbb{R} \to \mathbb{Z}$; for rational parameters it collapses to BISH,

witnessed by ${}_2F_1(\frac{1}{4}, \frac{1}{3}; \frac{2}{3}; z)$. GAGA descent is the unique CLASS obstruction, localized to Montel's theorem (sequential compactness in Cartan's Theorems A/B). The connection coefficients, despite being transcendental Gamma-function ratios, are BISH-computable — the classical taint enters through infinite-dimensional topology, not through transcendental evaluation.

Three broader conclusions follow. First, the ∞-categorical machinery used in the geometric Langlands proof (Lurie's higher algebra, ind-coherent sheaves, spectral decomposition) is logically inert: Kan extensions, colimits, and simplicial homotopy theory are BISH scaffolding that contributes zero omniscience cost. The logical expense of the geometric Langlands conjecture is inherited entirely from the RH correspondence's GAGA component. Second, Scholze's avoidance of Lurie's framework in condensed mathematics is explained by CRM as routing around the archimedean topology — the CLASS source identified by the Archimedean Obstruction Thesis — not as avoiding categorical complexity. Third, this audit establishes the baseline for a testable prediction: a condensed Riemann-Hilbert correspondence, if it eliminates GAGA dependence, would lower the logical cost from CLASS to WKL or below.

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